Exercise 4

For the Bachelor of Science course Digital Circuits.

1. Design a combinational circuit with three inputs and one output:
    a. The output is 1 when the binary value of the inputs is more than 2. The output is 0 otherwise.
    b. The output is 1 when the binary value of the inputs is not divisable by 3.
    c. The output is 1 when the binary value of the inputs is an even number.
2. Design a combinational circuit that converts a four-bit Gray code to a four-bit binary number. Implement the circuit with exclusive-OR gates.
3. Cnstruct a 16 × 1 multiplexer with two 8 × 1 and one 2 × 1 multiplexers. Use block diagrams.
4. An ABCD-to-seven-segment decoder is a combinational circuit that converts a decimal digit in BCD to an appropriate code for the selection of segments in an indicator used to display the decimal digit in a familiar form. The seven outputs of the decoder (a, b, c, d, e, f, g) select the corresponding segments in the display, as shown in *Figure 1*. Using a truth table and Karnaugh maps, design the BCD-to-seven-segment decoder using a minimum number of gates. The six invalid combinations should result in a blank display :

(a) Segment designation (b) Numerical designation for display

Figure 1

5. Implement the following Boolean function with a 8 × 1 multiplexer.

$$F(A, B, C, D) = \sum (0, 2, 5, 8, 10, 14)$$

6. Implemenet the following Boolean function with a 4 × 1 multiplexer and external gates.

$$F(A, B, C, D) = \sum (1, 3, 4, 11, 12, 13, 14, 15)$$